30 Years of Adaptive Neural Networks: Perceptron, Madaline, and Backpropagation

Bernard Widrow, Fellow, IEEE, and Michael A. Lehr
Information Systems Laboratory, Department of Electrical Engineering
Stanford University, Stanford, CA 94305-4055, USA

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Abstract

Fundamental developments in feedforward artificial neural networks from the past thirty years are reviewed. The central theme of this paper is a description of the history, origination, operating characteristics, and basic theory of several supervised neural network training algorithms including the Perceptron rule, the LMS algorithm, three Madaline rules, and the backpropagation technique. These methods were developed independently, but with the perspective of history they can all be related to each other. The concept underlying these algorithms is the "minimal disturbance principle," which suggests that during training it is advisable to inject new information into a network in a manner that disturbs stored information to the smallest extent possible.

1 Introduction

This year marks the 30th anniversary of the Perceptron rule and the LMS algorithm, two early rules for training adaptive elements. Both algorithms were first published in 1960. In the years following these discoveries, many new techniques have been developed in the field of neural networks, and the discipline is growing rapidly.

The first major extension of the feedforward neural network beyond Madaline I took place in 1971 when Werbos developed a backpropagation training algorithm which, in 1974, he first published in his doctoral dissertation. Unfortunately, Werbos's work remained almost unknown in the scientific community. In 1982, Parker rediscovered the technique and in 1985, published a report on it at M.I.T. Not long after Parker published his findings, Rumelhart, Hinton, and Williams also rediscovered the technique and, largely as a result of the clear framework within which they presented their ideas, they finally succeeded in making it widely known.

Section II explores fundamental concepts, Section III discusses adaptation and the minimal disturbance principle, Sections IV and V cover error correction rules, Sections VI and VII delve into steepest-descent rules, and Section VIII provides a summary.

2 Fundamental Concepts

2.1 The Adaptive Linear Combiner

The adaptive linear combiner receives at time k an input signal vector or input pattern vector $X_k = [x_{0k}, x_{1k}, x_{2k}, \ldots, x_{nk}]^T$ and a desired response d_k, a special input used to effect learning. The components of the input vector are weighted by a set of coefficients, the weight vector $W_k = [w_{0k}, w_{1k}, w_{2k}, \ldots, w_{nk}]^T$. The sum of the weighted inputs is then computed, producing a linear output, the inner product $s_k = X_k^T W_k$.

During the training process, input patterns and corresponding desired responses are presented to the linear combiner. An adaptation algorithm automatically adjusts the weights so that the output responses to the input patterns will be as close as possible to their respective desired responses. In signal processing applications, the most popular method for adapting the weights is the simple LMS (least mean square) algorithm, often called the Widrow-Hoff delta rule.

2.2 A Linear Classifier—The Single Threshold Element

The basic building block used in many neural networks is the "adaptive linear element," or Adaline. This is an adaptive threshold logic element. It consists of an adaptive linear combiner cascaded with a hard-limiting quantizer, which is used to produce a binary ± 1 output, $y_k = \text{sgn}(s_k)$. The bias weight w_{0k} which is connected to a constant input $x_0 = +1$, effectively controls the threshold level of the quantizer.

2.2.1 Linear Separability

With n binary inputs and one binary output, a single Adaline is capable of implementing certain logic functions. There are 2^n possible input patterns. A general logic implementation would be capable of classifying each pattern as either $+1$ or -1, in accord with the desired response. Thus, there are 2^{2^n} possible logic functions connecting n inputs to a single binary output. A single Adaline is capable of realizing only a small subset of these functions, known as the linearly separable logic functions or threshold logic functions.

The critical thresholding condition occurs when the linear output s equals zero:

$$s = x_1 w_1 + x_2 w_2 + w_0 = 0 \tag{1}$$

therefore

$$x_2 = -\frac{w_1}{w_2} x_1 - \frac{w_0}{w_2} \tag{2}$$

The three weights determine slope, intercept, and the side of the separating line that corresponds to a positive output. For Adalines with four weights, the separating boundary is a plane; with more than four weights, the boundary is a hyperplane.

2.2.2 Capacity of Linear Classifiers

In their 1964 dissertations, T. M. Cover and R. J. Brown both showed that the average number of random patterns with random binary desired responses that can be absorbed by an Adaline is approximately equal to twice the number of weights. This is the statistical pattern capacity C_s of the Adaline.

The deterministic pattern capacity C_d of the Adaline is $C_d = N_w$. An Adaline can learn any two-category pattern classification task involving no more patterns than that represented by its deterministic capacity.

$$P(\text{separable}) = \begin{cases} 1 & \text{for } N_p \leq N_w \\ \sum_{i=0}^{N_w - 1} \binom{N_p - 1}{i} \Big/ 2^{N_p - 1} & \text{for } N_p > N_w \end{cases} \tag{3}$$

3 Adaptation—The Minimal Disturbance Principle

The iterative algorithms described in this paper are all designed in accord with a single underlying principle. These techniques—the two LMS algorithms, Mays's rules, and the Perceptron procedure for training a single Adaline, the MRI rule for training the simple Madaline, as well as

MRII, MRIII, and backpropagation techniques for training multilayer Madalines—all rely upon the principle of minimal disturbance:

> *Adapt to reduce the output error for the current training pattern, with minimal disturbance to responses already learned.*

Unless this principle is practiced, it is difficult to simultaneously store the required pattern responses. The minimal disturbance principle is intuitive. It was the motivating idea that led to the discovery of the LMS algorithm and the Madaline rules.

4 Error Correction Rules—Single Threshold Element

4.1 Linear Rules

4.1.1 The α-LMS Algorithm

The α-LMS algorithm or Widrow-Hoff delta rule applied to the adaptation of a single Adaline embodies the minimal disturbance principle. The weight update equation for the original form of the algorithm can be written as:

$$\boxed{W_{k+1} = W_k + \alpha \frac{\epsilon_k}{|X_k|^2} X_k} \tag{4}$$

The time index or adaptation cycle number is k. W_{k+1} is the next value of the weight vector, W_k is the present value of the weight vector, and X_k is the present input pattern vector. The present linear error ϵ_k is defined to be the difference between the desired response d_k and the linear output $s_k = W_k^T X_k$ before adaptation:

$$\epsilon_k \triangleq d_k - W_k^T X_k \tag{5}$$

The choice of α controls stability and speed of convergence. For input pattern vectors independent over time, stability is ensured for most practical purposes if:

$$0 < \alpha < 2 \tag{6}$$

A practical range for α is $0.1 < \alpha < 1.0$.

4.2 Nonlinear Rules

4.2.1 The Perceptron Learning Rule

The Rosenblatt α-Perceptron processed input patterns with a first layer of sparse randomly connected fixed logic devices. The learning rule for the α-Perceptron is very similar to LMS, but its behavior is in fact quite different.

The adaptive threshold element of the α-Perceptron makes use of the "quantizer error" $\tilde{\epsilon}_k$, defined to be the difference between the desired response and the output of the quantizer:

$$\tilde{\epsilon}_k \triangleq d_k - y_k \tag{7}$$

The Perceptron rule does not adapt the weights if the output decision y_k is correct, that is, if $\tilde{\epsilon}_k = 0$. If the output decision disagrees with the binary desired response d_k, however, adaptation is effected by adding the input vector to the weight vector when the error $\tilde{\epsilon}_k$ is positive, or subtracting the input vector from the weight vector when the error $\tilde{\epsilon}_k$ is negative. Thus, half the product of the input vector and the quantizer error $\tilde{\epsilon}_k$ is added to the weight vector:

$$\boxed{W_{k+1} = W_k + \alpha \frac{\tilde{\epsilon}_k}{2} X_k} \tag{8}$$

The Perceptron rule is identical to the α-LMS algorithm, except that with the Perceptron rule, half of the quantizer error $\tilde{\epsilon}_k/2$ is used in place of the normalized linear error $\epsilon_k/|X_k|^2$ of the α-LMS rule. The Perceptron rule is nonlinear, in contrast to the LMS rule, which is linear.

The Perceptron rule has been proven to be capable of separating any linearly separable set of training patterns. If the training patterns are not linearly separable, the Perceptron algorithm goes on forever, and often does not yield a low-error solution, even if one exists.

5 Error Correction Rules—Multi-Element Networks

5.1 Madaline Rule I (MRI)

The MRI rule allows the adaptation of a first layer of hard-limited (signum) Adaline elements whose outputs provide inputs to a second layer, consisting of a single fixed-threshold-logic element which may be, for example, the OR gate, AND gate, or majority-vote-taker.

When adapting a given element, the weight vector can be moved in the LMS direction far enough to reverse the Adaline's output (absolute correction, or "fast" learning), or it can be adapted by the small increment determined by the α-LMS algorithm (statistical, or "slow" learning).

5.2 Madaline Rule II (MRII)

The MRI rule was extended to allow the adaptation of multilayer binary networks by Winter and Widrow with the introduction of Madaline Rule II (MRII). By the minimal disturbance principle, we select the first-layer Adaline with the smallest linear output magnitude and perform a "trial adaptation" by inverting its binary output. If the output Hamming error is reduced by this bit inversion, the weights of the selected Adaline element are changed by α-LMS in a direction collinear with the corresponding input vector.

6 Steepest-Descent Rules—Single Threshold Element

The method of steepest descent can be described by the relation:

$$W_{k+1} = W_k + \mu(-\nabla_k) \tag{9}$$

where μ is a parameter that controls stability and rate of convergence, and ∇_k is the value of the gradient at a point on the MSE surface corresponding to $W = W_k$.

6.1 The μ-LMS Algorithm

The μ-LMS algorithm works by performing approximate steepest descent on the MSE surface in weight space. Because it is a quadratic function of the weights, this surface is convex and has a unique (global) minimum. An instantaneous gradient based upon the square of the instantaneous linear error is:

$$\hat{\nabla}_k = \frac{\partial \epsilon_k^2}{\partial W_k} = -2\epsilon_k X_k \tag{10}$$

Making this replacement into the steepest descent equation yields:

$$\boxed{W_{k+1} = W_k + 2\mu\epsilon_k X_k} \tag{11}$$

This is the μ-LMS algorithm. The learning constant μ determines stability and convergence rate. For input patterns independent over time, convergence of the mean and variance of the weight vector is ensured for most practical purposes if:

$$0 < \mu < \frac{1}{\mathrm{tr}[R]} \tag{12}$$

where $\mathrm{tr}[R] = \sum(\text{diagonal elements of } R)$ is the average signal power of the X-vectors.

6.1.1 The Wiener Solution

The optimal weight vector W^*, generally called the Wiener weight vector, is obtained by setting the gradient to zero:

$$W^* = R^{-1}P \tag{13}$$

This is a matrix form of the Wiener-Hopf equation. The μ-LMS algorithm enables us to obtain an accurate estimate of W^* without first computing R^{-1} and P.

6.2 Backpropagation for the Sigmoid Adaline

Figure 18 shows a "sigmoid Adaline" element which incorporates a sigmoidal nonlinearity. The input-output relation of the sigmoid can be denoted by $y_k = \mathrm{sgm}(s_k)$. A typical sigmoid function is the hyperbolic tangent:

$$y_k = \tanh(s_k) \tag{14}$$

We shall adapt this Adaline with the objective of minimizing the mean square of the sigmoid error $\tilde{\epsilon}_k$, defined as:

$$\tilde{\epsilon}_k \triangleq d_k - y_k = d_k - \mathrm{sgm}(s_k) \tag{15}$$

The instantaneous gradient estimate obtained during presentation of the kth input vector X_k is:

$$\hat{\nabla}_k = \frac{\partial \tilde{\epsilon}_k^2}{\partial W_k} = -2\tilde{\epsilon}_k \cdot \mathrm{sgm}'(s_k) \cdot X_k \tag{16}$$

The algorithm is:

$$\boxed{W_{k+1} = W_k + 2\mu\tilde{\epsilon}_k \cdot \mathrm{sgm}'(s_k) \cdot X_k} \tag{17}$$

If the sigmoid is chosen to be the hyperbolic tangent function, then the derivative $\mathrm{sgm}'(s_k)$ is given by:

$$\mathrm{sgm}'(s_k) = \frac{\partial}{\partial s_k}\tanh(s_k) = 1 - \tanh^2(s_k) = 1 - y_k^2 \tag{18}$$

Accordingly, the backpropagation algorithm becomes:

$$\boxed{W_{k+1} = W_k + 2\mu\tilde{\epsilon}_k(1 - y_k^2)X_k} \tag{19}$$

6.3 Madaline Rule III for the Sigmoid Adaline

The Madaline Rule III (MRIII) algorithm circumvents the need for accurate realization of the sigmoid derivative. A small perturbation signal Δs is added to the sum s_k, and the effect of this perturbation upon output y_k and error $\tilde{\epsilon}_k$ is noted:

$$\frac{\Delta\tilde{\epsilon}_k}{\Delta s} \approx \frac{d\tilde{\epsilon}_k}{ds_k} \tag{20}$$

For small perturbations, MRIII and backpropagation are mathematically equivalent:

$$W_{k+1} = W_k + 2\mu\tilde{\epsilon}_k \cdot \mathrm{sgm}'(s_k) \cdot X_k \tag{21}$$

Thus, backpropagation and MRIII are mathematically equivalent if the perturbation Δs is small, but MRIII is robust, even with analog implementations.

6.4 MSE Surfaces of the Adaline

To demonstrate the nature of the square error surfaces associated with linear, sigmoid, and signum errors, a simple experiment with a two-input Adaline was performed. The MSE surfaces of $\mathbb{E}[(\epsilon)^2]$, $\mathbb{E}[(\tilde{\epsilon})^2]$, and $\mathbb{E}[(\hat{\epsilon})^2]$ are plotted as functions of the two weight values.

Only the linear error is guaranteed to have an MSE surface with a unique global minimum (assuming invertible R-matrix). The other MSE surfaces can have local optima. In nonlinear neural networks, gradient methods generally work better with sigmoid rather than signum nonlinearities.

7 Steepest-Descent Rules—Multi-Element Networks

7.1 Backpropagation for Networks

The publication of the backpropagation technique by Rumelhart et al. has unquestionably been the most influential development in the field of neural networks during the past decade.

When applied to multi-element networks, the backpropagation technique adjusts the weights in the direction opposite the instantaneous error gradient:

$$W_{k+1} = W_k - \mu \frac{\partial \mathcal{E}^2}{\partial W_k} \tag{22}$$

Now, however, W_k is a long m-component vector of all weights in the entire network. The instantaneous sum squared error $\mathcal{E}^2$ is the sum of the squares of the errors at each of the N_y outputs of the network:

$$\mathcal{E}^2 = \sum_{j=1}^{N_y} \tilde{\epsilon}_j^2 \tag{23}$$

In its simplest form, backpropagation training begins by presenting an input pattern vector X to the network, sweeping forward through the system to generate an output response vector Y, and computing the errors at each output. The next step involves sweeping the effects of the errors backward through the network to associate a "square error derivative" δ with each Adaline, computing a gradient from each δ, and finally updating the weights of each Adaline based upon the corresponding gradient.

8 Summary

This paper has reviewed fundamental concepts, learning algorithms, and applications of adaptive neural networks developed over the past thirty years. The minimal disturbance principle was shown to be the common thread linking the Perceptron rule, the LMS algorithm, Madaline rules, and backpropagation. All of these algorithms adapt the weights of a network to reduce output error for the current training pattern while minimally disturbing responses to previously learned patterns. The paper also discussed the relationship between these algorithms through the normalized training set concept, and showed that Madaline Rule III is mathematically equivalent to backpropagation.